—Sql Analysis—

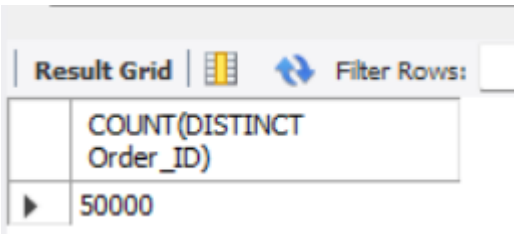
```sql
create database finance;
use finance;


-- Sales Performance

-- 1.How many total sales transactions/orders are there?


SELECT
    COUNT(DISTINCT Order_ID)
FROM
    sales;

-- The dataset contain 50k sales transactions.
```



```sql
-- 2.What is the total revenue generated from all sales?

SELECT
    ROUND(SUM(Net_Revenue), 2) AS Total_Revenue
FROM
    sales;


-- Total revenue generated is approximately 3355172364.39
```

Result Grid | Filter Rows:

	Total_Revenue
▶	3355172364.39

```sql
-- 3.What is the monthly revenue trend?
SELECT
    MONTH(Order_Date) AS Month,
    MONTHNAME(Order_Date) AS Month_Name,
    ROUND(SUM(Net_Revenue),2) AS Total_Revenue
FROM
    sales
GROUP BY MONTH(Order_Date) , MONTHNAME(Order_Date)
ORDER BY Month;

-- Revenue was highest in January and lowest in February.
-- After February monthly revenue remained relatively stable with moderate fluctuations.
```

Result Grid | Filter Rows:

	Month	Month_Name	Total_Revenue
▶	1	January	296949652.95
	2	February	256037879.88
	3	March	285881921.66
	4	April	284996068.2
	5	May	271896953.74
	6	June	280934191.21
	7	July	286929903.42

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	8	August	289800591.33
	9	September	272615226.71
	10	October	280638746.22
	11	November	272351377.7
	12	December	276139851.37

```sql
-- 4.Which region generates the highest revenue?

SELECT
    Region, ROUND(SUM(Net_Revenue),2) AS Total_Revenue
FROM
    sales
GROUP BY Region
ORDER BY Total_Revenue DESC
LIMIT 1;

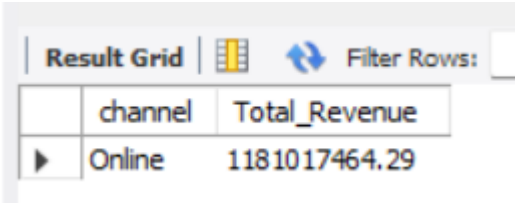
-- Central region has highest revenue approximately 735671057.76
```

Result Grid | Filter Rows:

	Region	Total_Revenue
▶	Central	735671057.76

```sql
-- 5.Which sales channel generates the highest revenue?
SELECT
    channel, ROUND(SUM(Net_Revenue), 2) Total_Revenue
FROM
    sales
GROUP BY Channel
ORDER BY Total_Revenue DESC
LIMIT 1;

-- Online channel has highest revenue approximately 1181017464.29
```

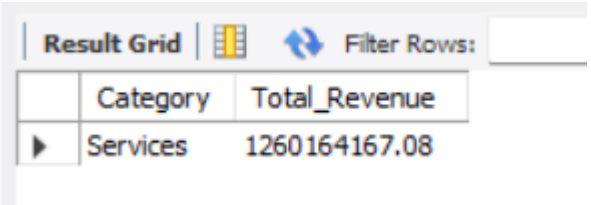


Result Grid | Filter Rows:

	channel	Total_Revenue
▶	Online	1181017464.29

```sql
-- 6.Which product category generates the highest revenue?
SELECT
    Category, ROUND(SUM(Net_Revenue),2) AS Total_Revenue
FROM
    sales
GROUP BY Category
ORDER BY Total_Revenue DESC
LIMIT 1;

-- Services category has highest revenue approximately 1260164167.08
```



Result Grid | Filter Rows:

	Category	Total_Revenue
▶	Services	1260164167.08

```sql
-- Profitability
-- 7.What is the total gross profit generated?
SELECT
    ROUND(SUM(Gross_Profit), 2) AS Total_Profit
FROM
    sales;

-- Total profit generated approximately 913898256.7
```

Result Grid

	Total_Profit
▶	913898256.7

```sql
-- 8.Which region generates the highest gross profit?
SELECT
    Region, ROUND(SUM(Gross_Profit),2) AS Total_Profit
FROM
    sales
GROUP BY Region
ORDER BY Total_Profit DESC LIMIT 1;

-- Central region generated the highest gross profit approximately 198813900.87
```

Result Grid Filter Rows:

	Region	Total_Profit
▶	Central	198813900.87

```sql
-- 9.What are the top 5 products by gross profit?

SELECT
    Product_ID, ROUND(SUM(Gross_Profit), 2) AS Total_Profit
FROM
    sales
GROUP BY Product_ID
ORDER BY Total_Profit DESC
LIMIT 5;

-- P025 has highest gross profit and P016 has lowest gross profit in top 5 product
```

Result Grid | Filter Rows:

Product_ID	Total_Profit
P025	91050294.69
P026	72655658.44
P002	70869255.91
P015	68554393.85
P016	64130356.59

```sql
-- 10.Which products generate high revenue but have a low profit margin?

-- Assumption:
-- High revenue = above the average product revenue
-- low profit margin =below 30%

SELECT
    Product_ID,
    ROUND(SUM(Net_Revenue),2) AS Total_Revenue,
    ROUND(SUM(Gross_Profit),2) AS Total_Gross_Profit,
    ROUND((SUM(Gross_Profit) / SUM(Net_Revenue)) * 100,2) AS Profit_Margin_pct
FROM
    sales
GROUP BY Product_ID
HAVING SUM(Net_Revenue) > (SELECT
        AVG(Product_Revenue)
    FROM
        (SELECT
            Product_ID, SUM(Net_Revenue) AS Product_Revenue
        FROM
            sales
        GROUP BY Product_ID) AS revenue_data)
    AND (SUM(Gross_Profit) / SUM(Net_Revenue)) * 100 < 30
ORDER BY Product_ID DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Product_ID	Total_Revenue	Total_Gross_Profit	Profit_Margin_pct
P028	227079419.96	33678846.14	14.83
P027	349210157.42	56079574.33	16.06
P026	355550801.81	72655658.44	20.43
P018	214166426.09	41536856.57	19.39
P013	112572225.16	23223527.62	20.63
P005	228120408.72	53279694.25	23.36
P002	253394355.36	70869255.91	27.97

	P001	170079128.09	23368399.89	13.74

```sql
-- 11. Which sales channel generates the highest gross profit?
SELECT
    Channel, ROUND(SUM(Gross_Profit), 2) AS Highest_Profit
FROM
    sales
GROUP BY Channel
ORDER BY Highest_Profit DESC
LIMIT 1;
-- Online channel has highest profit approximately 321923323.19
```

Result Grid | Filter Rows:

Channel	Highest_Profit
Online	321923323.19

```sql
-- 12. Who are the top 10 customers by revenue?
SELECT
    Customer_ID, ROUND(SUM(Net_Revenue),2) AS Total_Revenue
FROM
    sales
GROUP BY Customer_ID
ORDER BY Total_Revenue DESC
LIMIT 10;

--  Highest revenue generated from customer C0030 and lowest generated from C1554 in top 10 customers
```

Result Grid | Filter Rows:

Customer_ID	Total_Revenue
C0030	4590946.64
C1924	4092846.55
C1871	3917754.95
C1114	3838115.54
C1299	3806433.58
C0228	3626413.16
C0766	3530058.2

C0612	3505464.14
C0790	3375901.64
C1554	3346063.06

```sql
-- 13. Which region has the highest number of unique customers?
SELECT
    Region,
    COUNT(DISTINCT Customer_ID) AS Unique_Customers
FROM sales
GROUP BY Region
ORDER BY Unique_Customers DESC;

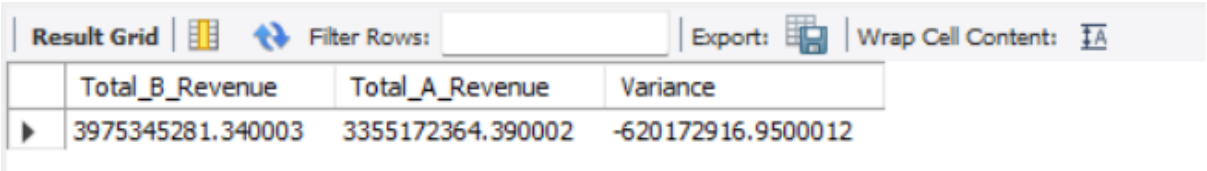
-- Central region has highest unique customers and west has lowest
```

Result Grid | Filter Rows:

Region	Unique_Customers
Central	443
South	402
East	391
North	384
West	380

```sql
-- 14. What is the variance between budgeted revenue and actual revenue?
SELECT
    SUM(b.Budget_Revenue) AS Total_B_Revenue,
    SUM(s.Actual_Revenue) AS Total_A_Revenue,
    SUM(s.Actual_Revenue) - SUM(b.Budget_Revenue) AS Variance
FROM
    (SELECT
        DATE_FORMAT(Date, '%Y-%m') AS Month,
            Product_ID,
            SUM(Budget_Revenue) AS Budget_Revenue
    FROM
        budget
    GROUP BY DATE_FORMAT(Date, '%Y-%m') , Product_ID) AS b
        JOIN
    (SELECT
        DATE_FORMAT(Order_Date, '%Y-%m') AS Month,
            Product_ID,
            SUM(Net_Revenue) AS Actual_Revenue
    FROM
        sales
    GROUP BY DATE_FORMAT(Order_Date, '%Y-%m') , Product_ID) AS s ON b.Month = s.Month
        AND b.Product_ID = s.Product_ID;

-- variance between budget revenue and actual revenue are approximately -620172916.9500012
```



Total_B_Revenue	Total_A_Revenue	Variance
3975345281.340003	3355172364.390002	-620172916.9500012

```sql
-- 15.Which months have actual revenue lower than budgeted revenue?
SELECT
    b.Month,
    b.Budget_Revenue,
    COALESCE(s.Actual_Revenue, 0) AS Actual_Revenue,
    COALESCE(s.Actual_Revenue, 0) - b.Budget_Revenue AS Variance
FROM
(
    SELECT
        DATE_FORMAT(Date, '%Y-%m') AS Month,
        SUM(Budget_Revenue) AS Budget_Revenue
    FROM budget
    GROUP BY DATE_FORMAT(Date, '%Y-%m')
) b
LEFT JOIN
(
    SELECT
        DATE_FORMAT(Order_Date, '%Y-%m') AS Month,
        SUM(Net_Revenue) AS Actual_Revenue
    FROM sales
    GROUP BY DATE_FORMAT(Order_Date, '%Y-%m')
) s
ON b.Month = s.Month
WHERE COALESCE(s.Actual_Revenue, 0) < b.Budget_Revenue

ORDER BY b.Month;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Month	Budget_Revenue	Actual_Revenue	Variance
2024-01	174501710.95	145668499.5999999	-28833211.350000083
2024-02	148414511.64000002	127751902.20000014	-20662609.43999988
2024-03	169348647.77999997	140802295.90000004	-28546351.879999936
2024-04	177454261.66	149760396.40000027	-27693865.259999722
2024-05	154942559.82999995	130385871.31000033	-24556688.519999623
2024-06	163380608.97000003	137305741.76000002	-26074867.21000001
2024-07	167508007.04999995	139188844.85999995	-28319162.189999998

2024-08	167141482.73	145506855.59999993	-21634627.130000055
2024-09	162877886.48	137216007.01000026	-25661879.46999973
2024-10	163298936.54	139504873.89000002	-23794062.649999976
2024-11	160295904.80999997	136898183.5299999	-23397721.28000006
2024-12	163630883.57	139781639.48000005	-23849244.089999944
2025-01	177559837.69000003	151281153.34999982	-26278684.340000212

2025-01	177559837.69000003	151281153.34999982	-26278684.340000212
2025-02	149443685.79000002	128285977.68000011	-21157708.10999991
2025-03	175223514.74000004	145079625.76000008	-30143888.97999996
2025-04	160085854.78	135235671.80000004	-24850182.97999996
2025-05	170414401.92000002	141511082.4300001	-28903319.48999992
2025-06	171984972.26999998	143628449.4500002	-28356522.819999784
2025-07	175144570.61999997	147741058.56	-27403512.059999973

2025-08	169068945.53999996	144293735.7299998	-24775209.81000015
2025-09	162418179.10999998	135399219.69999984	-27018959.410000145
2025-10	163540042.36999997	141133872.32999995	-22406170.04000002
2025-11	161476268.21000004	135453194.17000005	-26023074.03999999
2025-12	166189606.29	136358211.89	-29831394.400000006